Confidence envelopes for the false discoveries and False Discovery Rate with heterogeneous data

Romain Périer





Multiple testing setting

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Model and notations

Formal setting of multiple testing - notations

Let $X \in \mathcal{X}$ be a sample of distribution $P_0 \in \mathfrak{F}$ where $\mathfrak{F}$ is a set of probability measures. Define :

- $H_1, \ldots, H_m \subseteq \mathfrak{F}$, m null hypotheses,
- For all i, a p-value $p_i(X)$ or/and an e-value $e_i(X)$ such that :

$$\forall \alpha \in [0,1], \sup_{P \in H_i} \mathbb{P}_{X \sim P}(p_i \leq \alpha) \leq \alpha \text{ and } \sup_{P \in H_i} \mathbb{E}_{X \sim P}[e_i] \leq 1.$$

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Notations :

- $\mathcal{H}_0(P) = \{i : P \in H_i\}$,
- $m_0 = |\mathcal{H}_0(P)|$,
- $V(R) = |\mathcal{H}_0(P) \cap R|$,
- $\forall S \subseteq [\![1, m]\!], H_S = \bigcap_{i \in S} H_i$,
- $\mathbf{p} = (p_1, \dots, p_m)$, $\mathbf{e} = (e_1, \dots, e_m)$.

Multiple testing setting

Family-wise error rate and false discovery rate

Family-wise error rate (1/3)

Let R be a rejection set and $P \in \mathfrak{F}$. We define the Family-Wise Error Rate (FWER) as

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We say that:

- R achieves weak control of the FWER at level α if $\mathrm{FWER}_P(R) \leq \alpha$ for all $P \in H_{[\![1,m]\!]}$,
- R achieves strong control of the FWER at level α if $\mathrm{FWER}_P(R) \leq \alpha$ for all $P \in \mathfrak{F}$.

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Example 1 (Bonferroni procedure)

Let $\alpha \in [0,1]$ and $R_\alpha^{\mathrm{Bonf}}(\mathbf{p}) = \left\{ i \in [\![1,m]\!] : p_i \leq \frac{\alpha}{m} \right\}$. Then R_α^{Bonf} achieves strong control of the FWER at level α.

Family-wise error rate (2/3)

Denote by $\varphi = (\varphi_S)_{S \subseteq [\![1,m]\!]}$ a family of local tests. Then, the *closure principle* defines a new family of tests $\psi = (\psi_i)_{i \in [\![1,m]\!]}$, called *effective local tests*, as follows:

$$\forall S \subseteq [\![1,m]\!], \psi_S(X) = \min_{S \subseteq T \subseteq [\![1,m]\!]} \varphi_T(X). \tag{1}$$

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Proposition 1 (Closed testing procedure, Marcus et al., 1976)

Let $\alpha \in [0, 1]$ and assume that for all $P \in \mathfrak{F}$, $\varphi_{\mathcal{H}_0(P)}$ is an α-level test. Then, the multiple testing procedure R_φ defined by

$$R_\varphi(X) = \left\{ i \in [\![1, m]\!] : \psi_{\{i\}}(X) = 1 \right\}, \tag{2}$$

strongly controls the FWER *at level α.*

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Remark 1

If, for all $S \subseteq [\![1, m]\!]$, R_S is a rejection procedure that weakly controls the FWER *at level α for the submodel $(H_i)_{i \in S}$, then $\varphi_S = \mathbb{1}\left\{ |R_S| > 0 \right\}$ defines a family of α-level local tests.*

Family-wise error rate (3/3)

Example 2

The Holm procedure at level α is the closure of the Bonferroni procedure at level α and is defined by,

$$R_\alpha^{\text{Holm}} := \left\{ i \in [\![1, m]\!] : p_i(X) \leq \frac{\alpha}{m - k^* + 1} \right\}, \tag{3}$$

where $k^* = \max\left\{ k \in [\![1, m]\!] : \forall i \leq k, p_{(i)}(X) \leq \frac{\alpha}{m-i+1} \right\} \cup \{0\}$.

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For all $\alpha \in [0, 1]$, the Simes procedure at level α is defined by

$$R_\alpha^{\text{Sim}}(\mathbf{p}) = \left\{ i \in [\![1, m]\!] : p_i(X) \leq \frac{\alpha k^*}{m} \right\}, \tag{4}$$

where $k^* = \max\left\{ k \in [\![1, m]\!] : p_{(k)}(X) \leq \frac{\alpha k}{m} \right\} \cup \{0\}$. It weakly controls the FWER at level α under the assumption of independence of the p-values.

False Discovery Rate

We define the False Discovery Rate (FDR) as the mean of the False Discovery Proportion (FDP):

$$\mathrm{FDP}(R) = \frac{V(R)}{|R| \vee 1}, \quad \mathrm{FDR}_P(R) = \mathbb{E}_{X \sim P}[\mathrm{FDP}(R)].$$

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Example 3 (Benjamini-Hochberg procedure, Benjamini and Hochberg, 1995)

Let $\alpha \in [0,1]$ and $P \sim \mathfrak{F}$. Then, if $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p-values and is independent of $(p_i)_{i \in \mathcal{H}_1}$, $\mathrm{FDR}_P\left(R_\alpha^{\mathrm{Sim}}\right) \leq \alpha$.

Example 4 (Benjamini-Yekutieli procedure, Benjamini and Yekutieli, 2001)

Let $\alpha \in [0,1]$ and $P \sim \mathfrak{F}$. Then, $\mathrm{FDR}_P\left(R_{\alpha/h_m}^{\mathrm{BH}}\right) \leq \alpha$, where $h_m = \sum_{i=1}^m 1/i$.

Multiple testing setting

Heterogeneous multiple testing setting

Examples of heterogeneous MT settings

Binomial tests : Assume $X = (X_1, \dots, X_m)$, where $(X_i)_{1 \leq i \leq m}$ are independent variables such that $X_i \sim \mathcal{B}(n_i, \pi_i)$ under H_i. Then :

- $p_i(X_i) := \mathbb{P}_{Y \sim \mathcal{B}(n_i, \pi_i)}(Y \leq X_i) = \sum_{j=0}^{X_i} \binom{n_i}{j} \pi_i^j (1 - \pi_i)^{n_i - j}$ is a p-value for i,
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- Fisher's exact tests (see Agresti, 2013),
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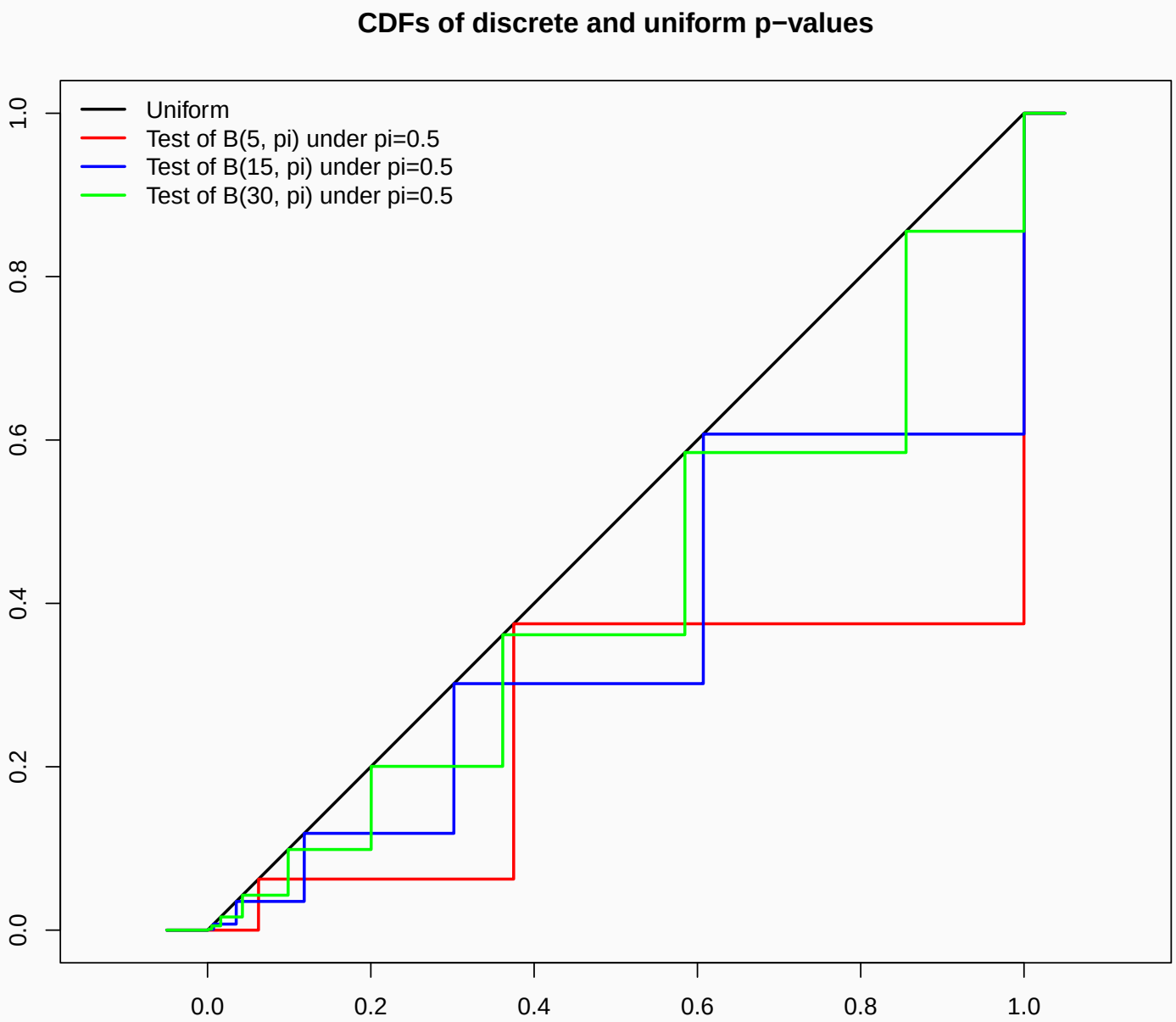
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Assumption 1 (Stochastic heterogeneous lower bound)

For each $i \in [\![1, m]\!]$, there exists a known nondecreasing function F_i such that, if $P_0 \in H_i$, the cdf of p_i is upper bounded (as a function) by F_i. Mathematically,

$$\forall P \in \mathfrak{F}, \forall i \in \mathcal{H}_0(P), \forall x \in \mathbb{R}, \ \mathbb{P}_{X \sim P}\big(p_i(X) \leq x\big) \leq F_i(x).$$

Cdf of discrete p-values



Confidence envelopes with heterogeneous data (Périer et al., 2026)

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Confidence envelopes and JER

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Definition 1 (Confidence envelope)

Let $\alpha \in (0,1)$. A confidence envelope of level α is a statistic $\hat{V}_\alpha : \mathcal{P}(\mathbb{N}_m) \to \mathbb{N}$ such that, for all $P \in \mathfrak{F}$,

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Definition 2 (Joint Error Rate)

For a reference family $\mathfrak{R} = \big((R_k, \zeta_k)\big)_{1\leq k\leq K}$ and $P \in \mathfrak{F}$, the joint error rate is :

$$JER(\mathfrak{R}, P) := \mathbb{P}_{X\sim P}\Big(\exists k \in \mathbb{N}_K : V(R_k) > \zeta_k\Big).$$

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In the following, we will be in a **top-k setting** :

$$R_k = \Big\{\sigma(i), 1 \leq i \leq k\Big\} \text{ (where } p_{\sigma(1)} \leq \cdots \leq p_{\sigma(m)}\text{)}.$$

Confidence envelopes with heterogeneous data (Périer et al., 2026)

Bretagnolle's calibration

DKWM and Bretagnolle inequalities

Proposition 2 (DKWM inequality - Improved version from Reeve, 2024)

Let $X_1, \dots, X_m \overset{iid}{\sim} F$ *and* $\hat{F}_m(t) := \frac{1}{m} \sum_{1 \leq i \leq m} \mathbb{1}_{X_i \leq t}$ *for all* $t \in \mathbb{S}$.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{m} \sup_{t \in \mathbb{R}} \left(\hat{F}_m - F \right)(t) > \lambda \right) \leq \exp\left(-2\lambda^2 \right).$$

When $(p_i)_{1 \leq i \leq m} \overset{iid}{\sim} \mathcal{U}\left([0,1]\right)$, $F(t) = t$.

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Proposition 3 (Bretagnolle's inequality, Shorack and Wellner, 1986 with the improvement from Reeve, 2024)

Let $X_1, \dots, X_m$ *be independent, denote by* F_i *the cdf of* X_i *and* $\bar{F}$ *the mean of* $F_i's$.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{m} \sup_{t \in \mathbb{R}} \left(\hat{F}_m - \bar{F} \right)(t) > \lambda \right) \leq e \cdot \exp\left(-2\lambda^2 \right).$$

Non-adaptive top-k bound

Theorem 1 (Bretagnolle's top-k calibration)

Let $\alpha \in (0, 1]$ *and* $\lambda_\alpha = \sqrt{\frac{1+\log(1/\alpha)}{2}}$. *Assume that* $(p_i)_{i \in \mathcal{H}_0}$ *is an independent family of p-values. Define*

$$\zeta_k = k \wedge \left\lfloor m\bar{F}\left(p_{(k)}\right) + \sqrt{m}\lambda_\alpha \right\rfloor.$$

Then, the reference family $\mathfrak{R} = \left((R_k, \zeta_k)\right)_{k \in \mathcal{K}}$ *has its* JER *bounded by* α.

Adaptive top-k bound

Theorem 2 (Adaptive version of Bretagnolle's top-k calibration)

Let $\alpha \in (0,1]$ and $\mathbb{G}(t) = \left(\mathbb{1}_{\{p_i \leq t\}} - F_i(t)\right)_{1 \leq i \leq m}$. Assume that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p-values. Let $\hat{m}_0$ such that

$$\left\{ \sup_{t \in [0,1]} \sum_{i \in \mathcal{H}_0} \mathbb{G}_i(t) \leq \sqrt{m_0} \lambda_\alpha \right\} \subset \{ m_0 \leq \hat{m}_0 \}, \text{ and}$$

$$\zeta_k = k \wedge \left\lfloor \sum_{1 \leq i \leq \hat{m}_0} \left(\mathbf{F}\left(p_{(k)}\right) \right)_{[i]} + \sqrt{\hat{m}_0} \lambda_\alpha \right\rfloor^{1}.$$

Then, the reference family $\mathfrak{R} = \left((R_k, \zeta_k)\right)_{k \in \mathcal{K}}$ has its JER *bounded by α.*

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Sketch of proof:

- $|R_k \cap \mathcal{H}_0| \leq \sum_{i \in \mathcal{H}_0} \mathbb{1}_{p_i \leq p_{(k)}}$,
- Evaluate Bretagnolle's inequality at point $p_{(k)}$,
- Bound $\sqrt{m_0}$ by $\sqrt{\hat{m}_0}$ and $\sum_{i \in \mathcal{H}_0} F_i\left(p_{(k)}\right)$ by $\sum_{1 \leq i \leq \hat{m}_0} \left(\mathbf{F}\left(p_{(k)}\right) \right)_{[i]}$ (worst case).

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Confidence envelopes with heterogeneous data (Périer et al., 2026)

Inversion procedure and shortcuts

Inversion procedure

Recall, for $A \subseteq \{1, \ldots, m\}$, the definition of the intersection hypothesis: $H_A := \bigcap_{i \in A} H_i$.

Let $\varphi = \left(\varphi_A\right)_{A \subseteq \{1,\ldots,m\}}$ be a family of local tests. The inversion procedure is based on the set of potential candidates for the true nulls $\mathcal{H}_0(P)$:

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- **Advantage :** Best bound on $V(S)$ from local tests,
- **Drawback :** Exponential computational cost.

Inversion procedure and concentration inequalities (1/2)

Suppose that for all $A \subseteq \{1, \dots, m\}$, $\varphi_A = \sup_{t \in [0,1]} \varphi_{A,t}$.

Example 5

Let $\alpha \in (0, 1]$ and $A \subseteq [\![1, m]\!]$. From Bretagnolle's inequality,

$$\varphi_{A,t} = \mathbb{1}\left\{ \sqrt{|A|}\big(\hat{F}_A - \bar{F}_A\big)(t) > \lambda_\alpha \right\} = \mathbb{1}\left\{ \sum_{i \in A} \mathbb{G}_i(t) > \sqrt{|A|}\lambda_\alpha \right\},$$

where $\mathbb{G}(t) = \big(\mathbb{1}_{\{p_i \leq t\}} - F_i(t)\big)_{1 \leq i \leq m}$, is a local test of level α.

Inversion procedure and concentration inequalities (1/2)

Suppose that for all $A \subseteq \{1, \dots, m\}$, $\varphi_A = \sup_{t \in [0,1]} \varphi_{A,t}$.

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Lemma 1

For all $S \subseteq \mathbb{N}_m$,

$$\hat{V}_{\mathcal{N}}(S) = \max\left\{ \ell \in \{0, \dots, |S|\} : \min_{A : |A \cap S| = \ell} \sup_{t \in [0,1]} \varphi_{A,t} = 0 \right\}.$$

Inversion procedure and concentration inequalities (2/2)

Proposition 5

Denote by $\varphi^{SC}_{S,\ell} := \max_{t\in[0,1]} \min_{A:|A\cap S|=\ell} \varphi_{A,t}$, *then :*

$$\hat{V}_{\mathcal{N}}(S) \leq \max\left\{\ell \in \{0,\dots,|S|\} : \varphi^{SC}_{S,\ell} = 0\right\}.$$

Inversion procedure and concentration inequalities (2/2)

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Example 6

With Bretagnolle's local test, for $t \in [0,1]$,

$$\min_{A:|A\cap S|=\ell} \varphi_{A,t} =$$

$$\min_{0\leq j\leq m-|S|} \mathbb{1}\left\{\sum_{1\leq i\leq \ell} \big(\mathbb{G}(t)\big)_{(S:i)} + \sum_{1\leq i\leq j} \big(\mathbb{G}(t)\big)_{(S^c:i)} > \lambda_\alpha \sqrt{\ell + j}\right\}.$$

- Reduces computation cost to $\mathcal{O}\big(m^2 \log(m)\big)$.

Upper bound on m_0 via Bretagnolle's inequality

Theorem 3 (Bretagnolle's upper bound on m_0)

Let $\alpha \in (0,1]$. Suppose that for all $k \in \mathcal{K}$, $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p-values.
Define $\lambda_\alpha = \sqrt{\frac{1+\log(1/\alpha)}{2}}$.
Then, within the event $\{\mathcal{H}_0 \in \mathcal{N}\}$, we have

$$m_0 \leq \max\left\{ \ell \in [\![0, m]\!] : \forall t \in [0,1], \sum_{1 \leq j \leq \ell} \left(\mathbb{G}(t)\right)_{(j)} \leq \lambda_\alpha \sqrt{\ell} \right\} := \hat{m}_0.$$

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Sketch of proof :

- Use Example 6 with $S = \{1, \ldots, m\}$,
- Apply Lemma 5, second inequality.

Confidence envelopes with heterogeneous data (Périer et al., 2026)

Adaptive calibration

Adaptive version of Bretagnolle's calibration

Theorem 4 (Adaptive version of Bretagnolle's top-k calibration)

Let $\alpha \in (0, 1]$. Suppose that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p-values. Define $\lambda_\alpha = \sqrt{\frac{1+\log(1/\alpha)}{2}}$, $\mathbf{F}(t) := \big(F_i(t)\big)_{1 \le i \le m}$, $\hat{m}_0$ as the estimator of m_0 from Theorem 3 and

$$\zeta_k = k \wedge \left\lfloor \sum_{1 \le i \le \hat{m}_0} \Big(\mathbf{F}\big(p_{(k)}\big)\Big)_{[i]} + \sqrt{\hat{m}_0}\lambda_\alpha \right\rfloor^2.$$

Then, the reference family $\mathfrak{R} = (R_k, \zeta_k)_{1 \le k \le m}$ has its JER *bounded by α.*

[2] $x_{[i]}$ is the i-th largest coordinate of x.

Simulations (1/2)

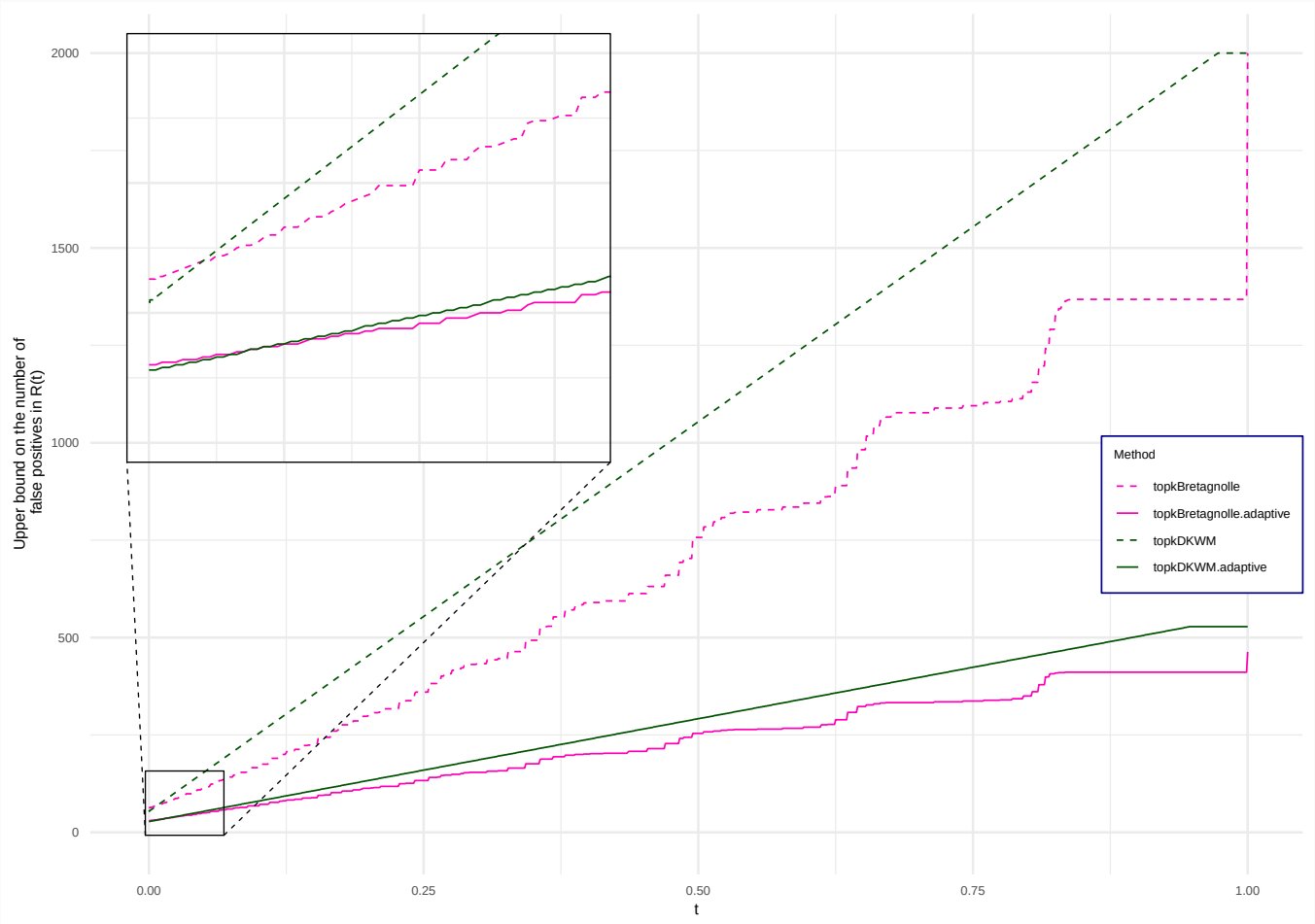


Figure 2: Upper bounds on the number of false positives in the set $R(t)$. $\pi_0 = 0.2$ and $q = 0.4$.

Simulations (2/2)

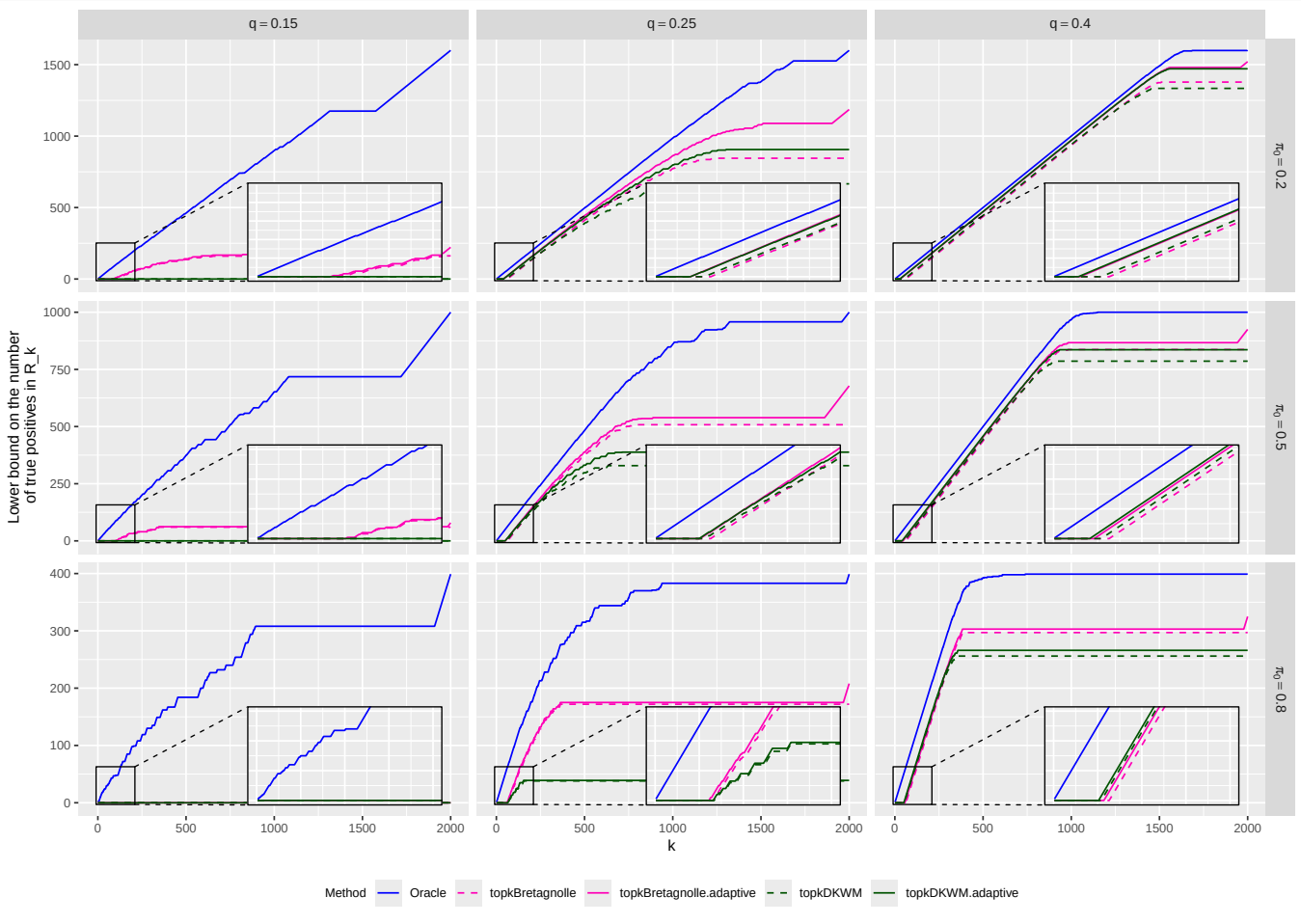


Figure 3: Lower bounds on the number of true positives in the set R_k.

Confidence envelopes with heterogeneous data (Périer et al., 2026)

Comparison with existing literature

Bound efficiency comparisons

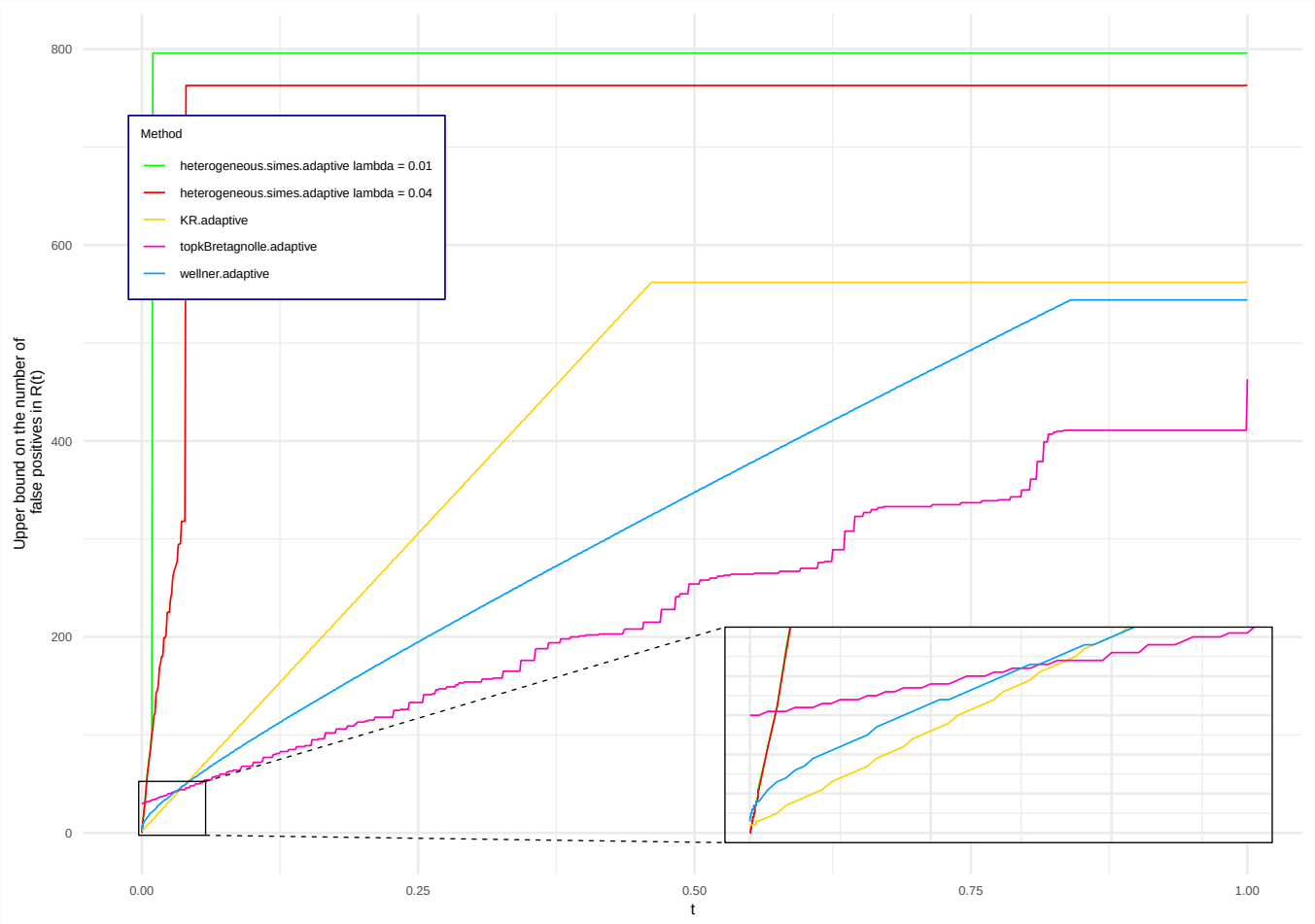


Figure 4: Upper bounds on the number of false positives in the set $R(t)$. $\pi_0 = 0.2$ and $q = 0.4$.

Envelope comparisons

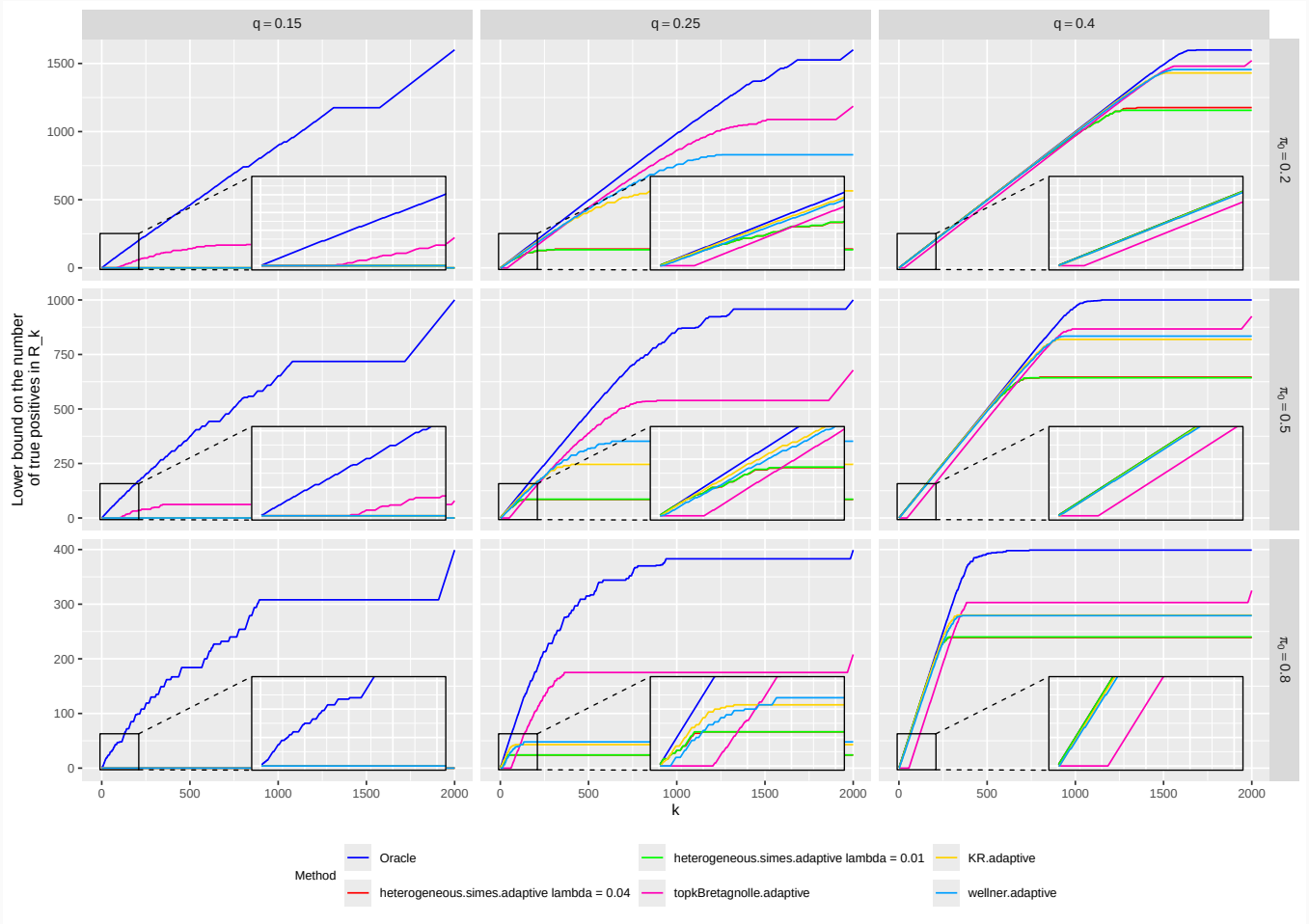


Figure 5: Lower bounds on the number of true positives in the set R_k.

e-Closure Principle with heterogeneous data (to appear)

e-Closure Principle with heterogeneous data (to appear)

Background on e-values

Background on e-values (1/2)

Definition 3 (p-to-e calibrator, Vovk and Wang, 2021)

A nonincreasing function $g : [0, 1] \to [0, +\infty]$ *is a p-to-e calibrator if* $\int_0^1 g \, \mathrm{d}\lambda \leq 1$, *where* λ *is the Lebesgue measure.*

Background on e-values (1/2)

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Example 7 (BY calibrator, Xu et al., 2024)

Let $\alpha \in \mathbb{R}^*$, $s \in \mathbb{N}$. Define $h_s = \sum_{i=1}^{s} \frac{1}{i}$ and g_s^α, for all $x \in [0,1]$, by,

$$g_s^\alpha(x) = \frac{s}{\alpha} \frac{\mathbb{1}\{h_s x \leq \alpha\}}{\lceil s h_s x/\alpha \rceil \vee 1}.$$

Background on e-values (2/2)

Definition 4 (Compound p-values, Ignatiadis et al., 2025)

Let $p_1, \ldots, p_m$ be $[0,1]$-valued random variables. Then $p_1, \ldots, p_m$ are compound p-values if the following holds :

$$\forall t \in [0,1], \ \sup_{P \in \mathfrak{F}} \sum_{i \in \mathcal{H}_0(P)} \mathbb{P}_{X \sim P}(p_i \leq t) \leq mt.$$

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Let $e_1, \ldots, e_m$ be $[0,\infty]$-valued random variables. Then $e_1, \ldots, e_m$ are compound e-values if the following holds :

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Background on e-values (2/2)

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$$\sup_{P \in \mathfrak{F}} \sum_{i \in \mathcal{H}_0(P)} \mathbb{E}_P[e_i] \leq m.$$

Proposition 6 (Ignatiadis et al., 2025)

Let $p_1, \ldots, p_m$ be compound p-values and g be a calibrator. Then $g(p_1), \ldots, g(p_m)$ are compound e-values.

e-Closure Principle with heterogeneous data (to appear)

e-Closure principle

e-Closure principle

Definition 6 (e-Closure principle, Xu et al., 2026)

Let $\mathbf{e} = (e_S)_{S \subseteq [\![1,m]\!]}$ *be a family of local e-values. Then, the e-Closure Principle defines a set of procedures by the following:*

$$\mathcal{R}_\alpha(\mathbf{e}) = \left\{ R \subseteq [\![1,m]\!] : \alpha e_S \geq \frac{|R \cap S|}{|R| \vee 1} \ \forall S \subseteq [\![1,m]\!] \right\}.$$

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Proposition 7 (Simultaneous control of the FDR, Xu et al., 2026)

For all $P \in \mathfrak{F}$, $\mathbb{E}_P \left[\max_{R \in \mathcal{R}_\alpha(\mathbf{e})} \frac{|R \cap \mathcal{H}_0(P)|}{|R| \vee 1} \right] \leq \alpha$*; hence for all random set* $R \subseteq [\![1,m]\!]$ *such that* $R \in \mathcal{R}_\alpha(\mathbf{e})$ *a.s.,* R *has its* FDR *upper-bounded by* α*.*

e-Closure principle

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In Xu et al., 2026, the authors used the e-Closure principle with the choice $e_S^{\mathrm{BY}} = \frac{1}{|S|} \sum_{i \in S} g^\alpha_{|S|}(p_i)$ to derive a more powerful procedure than BY that controls the FDR at level α.

e-Closure Principle with heterogeneous data (to appear)

Heterogeneous e-Closure principle

Heterogeneous S-locally compound p-values

Proposition 8

Assume Assumption 1 holds and let $S \subseteq [\![1, m]\!]$. *Define*

$$\bar{F}^S = \frac{1}{|S|} \sum_{i \in S} F_i.$$

Then $\left(\bar{F}^S(p_i)\right)_{i \in S}$ *are* S*-locally compound p-values, i.e., compound p-values for the sub multiple testing problem restricted to* S.

Heterogeneous S-locally compound p-values

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Then $\left(\bar{F}^S(p_i)\right)_{i \in S}$ are S-locally compound p-values, i.e., compound p-values for the sub multiple testing problem restricted to S.

The family $\left(\bar{F}(p_i)\right)_{i \in [\![1,m]\!]}$ was already used in Döhler (2018) to derive the DBY procedure, which is defined by $R_\alpha^{\mathrm{DBY}}(\mathbf{p}) = R_\alpha^{\mathrm{BY}}\left(\bar{F}(\mathbf{p})\right)$, and the Heyse procedure Heyse (2011), which is defined by $R_\alpha^{H}(\mathbf{p}) = R_\alpha^{\mathrm{BH}}\left(\bar{F}(\mathbf{p})\right)$.

Heterogeneous e-Closure principle

For $\alpha \in (0,1]$ fixed, we define for all $S \subseteq [\![1,m]\!]$ and $i \in S$,

$$e_{S,i}^{\mathrm{HBY}} := g_{|S|}^{\alpha}\left(\bar{F}^{S}(p_i)\right), \text{ and } e_{S,i}^{\mathrm{DBY}} := g_{m}^{\alpha m/|S|}\left(\bar{F}^{S}(p_i)\right). \tag{5}$$

Then, we define for all $S \subseteq [\![1,m]\!]$,

$$e_{S}^{\mathrm{HBY}} = \frac{1}{|S|}\sum_{i\in S} e_{S,i}^{\mathrm{HBY}}, \text{ and } e_{S}^{\mathrm{DBY}} = \frac{1}{|S|}\sum_{i\in S} e_{S,i}^{\mathrm{DBY}}. \tag{6}$$

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Theorem 5

Assume Assumption 1 holds. Then:

- $\mathcal{R}_{\alpha}\left(\mathbf{e}^{\mathrm{BY}}\right) \subseteq \mathcal{R}_{\alpha}\left(\mathbf{e}^{\mathrm{HBY}}\right)$, *and*
- $R_{\alpha}^{\mathrm{DBY}} \in \mathcal{R}_{\alpha}\left(\mathbf{e}^{\mathrm{DBY}}\right)$.

Shortcut for the heterogeneous e-Closure principle

For all $R \subseteq [\![1, m]\!]$, we define the following shortcut:

$$\mathcal{R}^{\mathsf{SC}}_{\alpha,g}(\mathbf{p}) = \left\{ R \subseteq [\![1, m]\!] : \alpha e_{S,R,g} \geq \frac{|R \cap S|}{|R| \vee 1}, \forall S \subseteq [\![1, m]\!] \right\}, \text{ where}$$

$$e_{S,R,g} = \frac{1}{|S|} \sum_{i \in S} g\left(\frac{1}{|S|} \left(\sum_{1 \leq j \leq |S \cap R|} \left(\mathbf{F}(p_i)\right)_{[j:R]} + \sum_{1 \leq j \leq |S \cap R^c|} \left(\mathbf{F}(p_i)\right)_{[j:R^c]} \right) \right).$$

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Proposition 9

For all p-to-e calibrator g,

$$\mathcal{R}_\alpha\left(\left(\frac{1}{|S|} \sum_{i \in S} g(p_i) \right)_{S \subseteq [\![1,m]\!]} \right) \subseteq \mathcal{R}^{\mathrm{SC}}_{\alpha,g}(\mathbf{p}) \subseteq \mathcal{R}_\alpha\left(\left(\frac{1}{|S|} \sum_{i \in S} g\left(\bar{F}(p_i)\right) \right)_{S \subseteq [\![1,m]\!]} \right).$$

Simulation study (1/4)

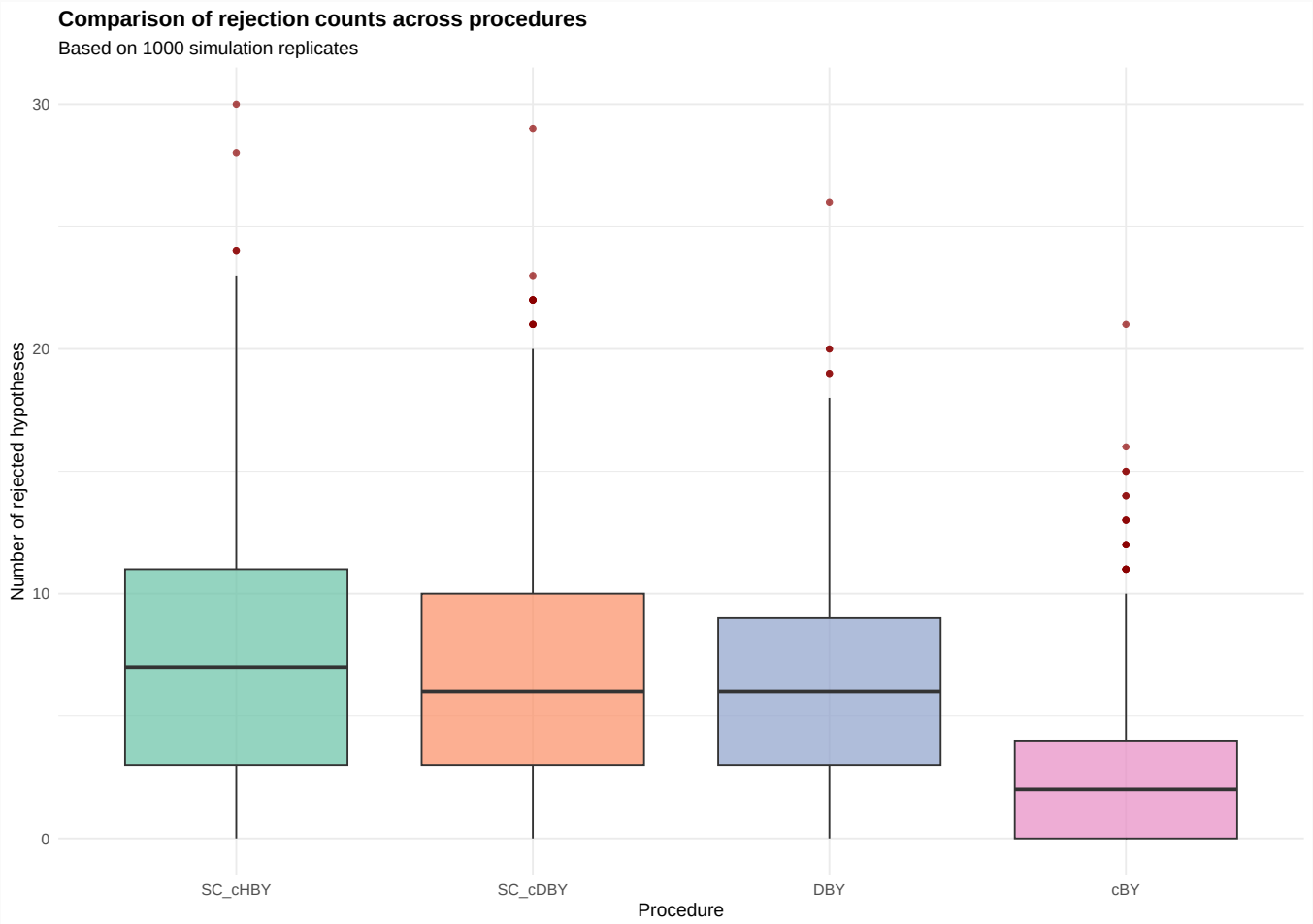


Figure 6: Boxplot of the number of rejections for the different procedures ($q = 0.25, m = 100$).

Simulation study (2/4)

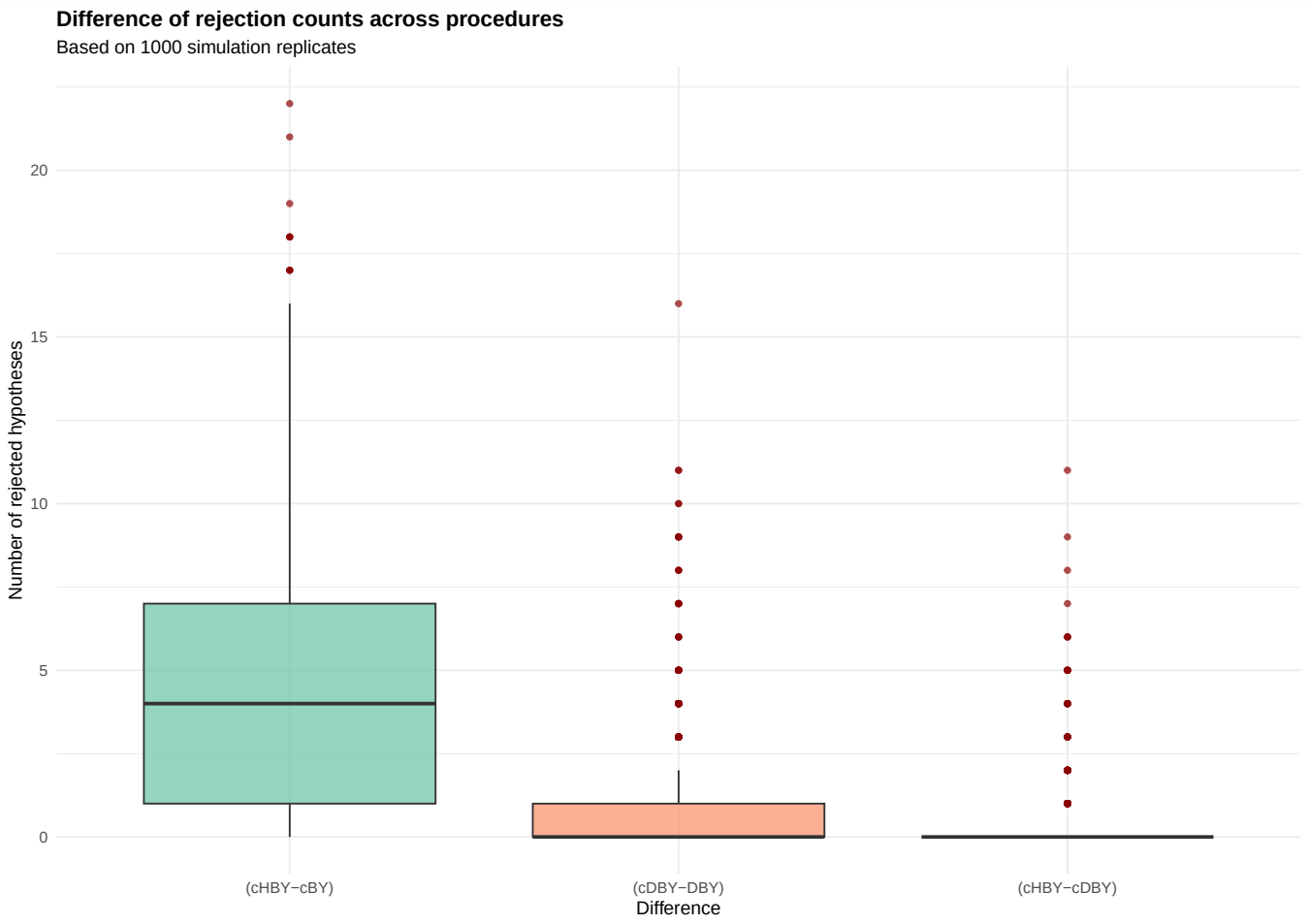


Figure 7: Boxplot of differences between the number of rejections.

Simulation study (3/4)

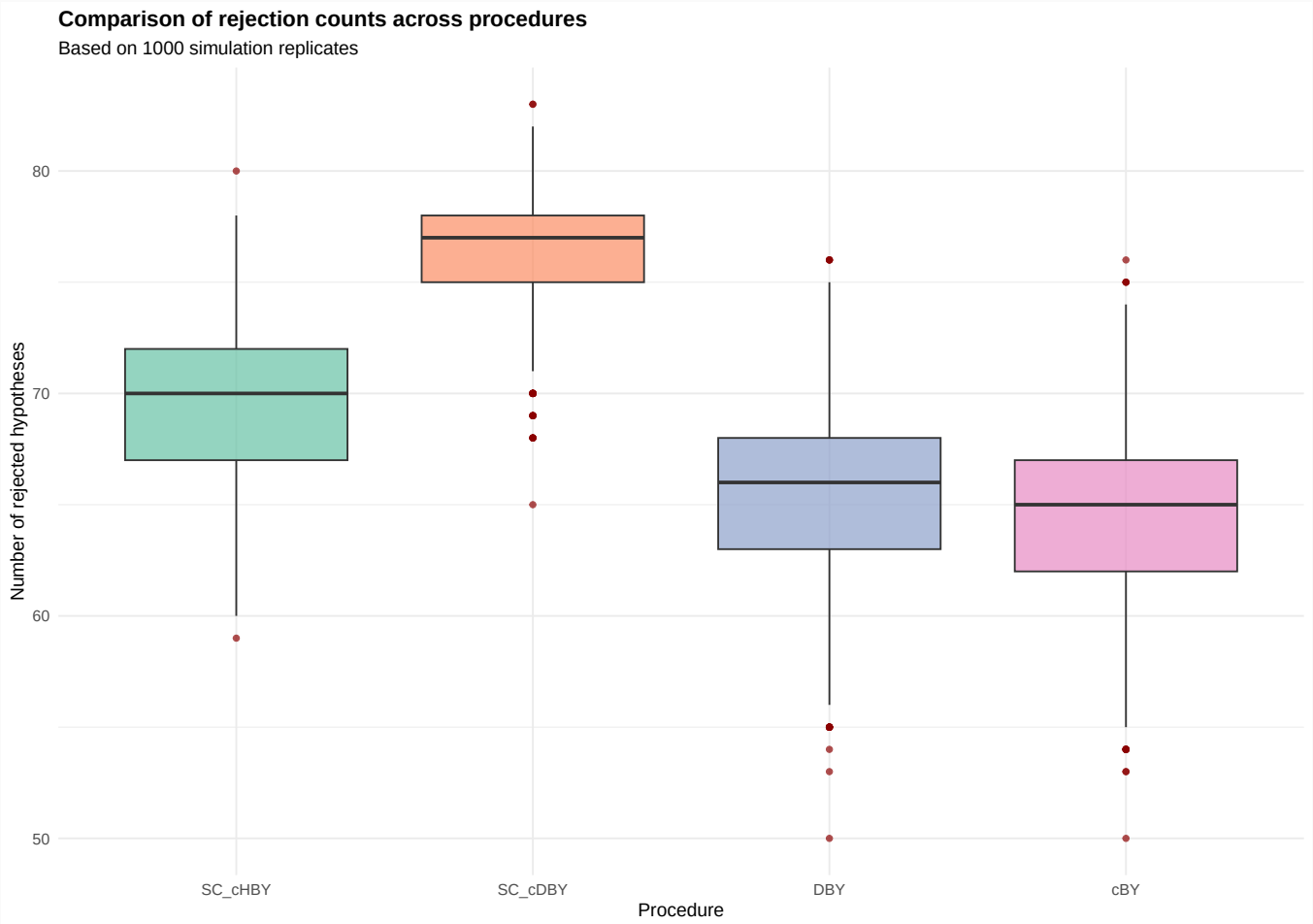


Figure 8: Boxplot of the number of rejections for the different procedures ($q = 0.4, m = 100$).

Simulation study (4/4)

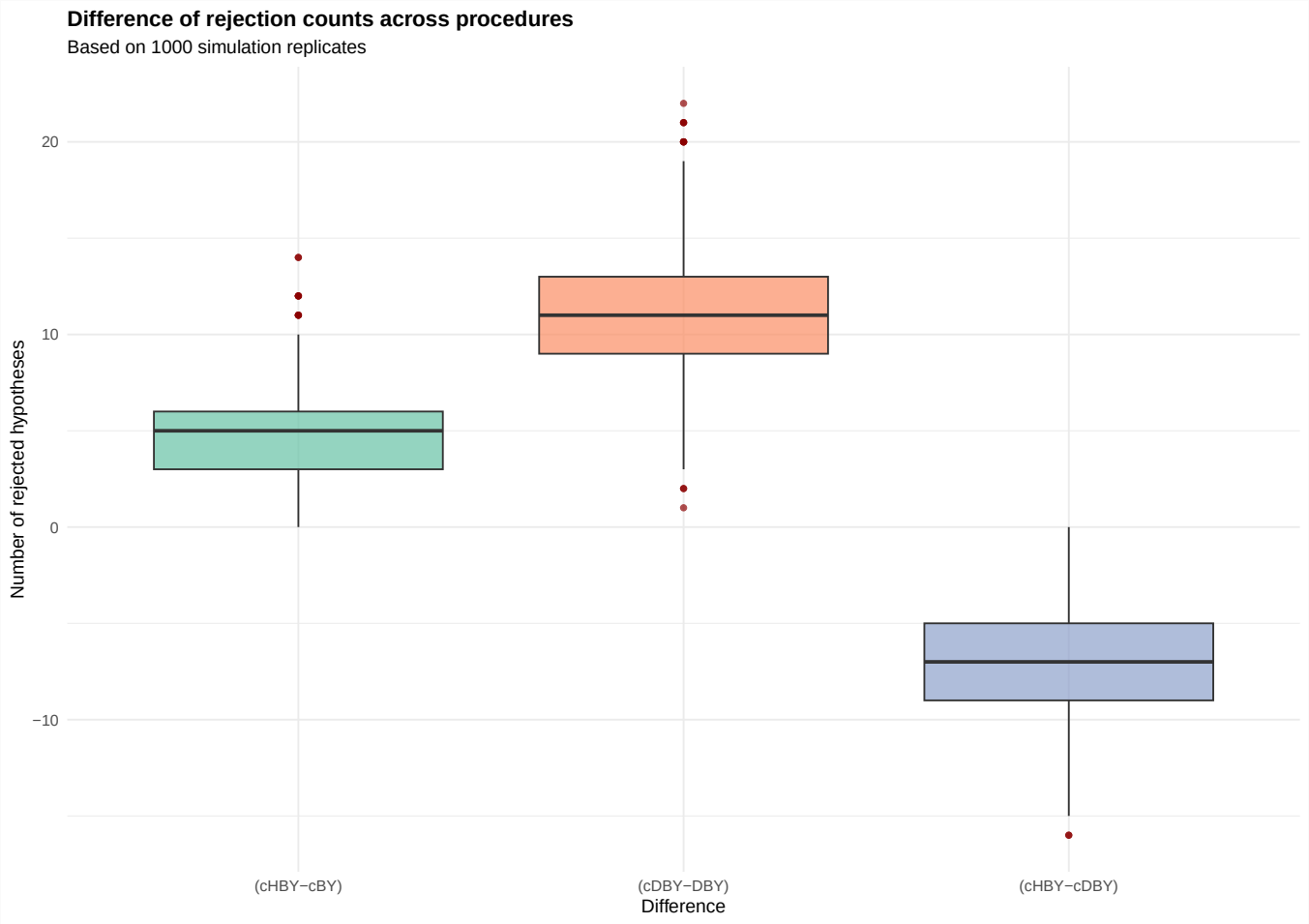


Figure 9: Boxplot of differences between the number of rejections.

Summary and perspectives

Summary :

- Considered heterogeneity in the context of confidence envelopes and FDR control,
- Provided computable (inexact) shortcuts.

Perspectives :

- Use the results of (Barber and Samworth, 2025) to derive new heterogeneous confidence bounds,
- Provide a version of Bretagnolle's inequality without the "$exp(1)$" factor thanks to the recent works from Reeve (2024) or Boudec (2026).

Thanks a lot for your attention!

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Appendix: Confidence envelopes with heterogeneous data

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Precisions on interpolated bounds

Interpolated confidence envelope

Proposition 10 (Interpolated bound, Blanchard et al., 2020)

Let $\mathfrak{R} = \big((R_k, \zeta_k)\big)_{1 \le k \le K}$ *a family with* JER *bounded by* $\alpha \in [0,1]$ *and* $\mathcal{A}(\mathcal{R}) := \big\{A \subset \mathbb{N}_m : \forall k \in \mathbb{N}_K, |R_k \cap A| \le \zeta_k\big\}$.
Then $V^*_{\mathfrak{R}}(S) := \max_{A \in \mathcal{A}(\mathcal{R})} |R \cap A|$ *is a level* α *post hoc bound.*

Proposition 11 (Interpolated bound nested setting)

Assume $(R_k)_k$ *is a family of regions such that*

$$\forall k \in \{1, \dots, K\}, R_k \subset R_{k+1}.$$

Then:

$$V^*_{\mathfrak{R}}(S) = \min_{1 \le k \le K} \big(\zeta_k + |S \setminus R_k|\big) \wedge |S|.$$

Appendix: Confidence envelopes with heterogeneous data

Complements on Uniform Wellner and K& R inequalities

Uniform Wellner Meah et al., 2024 and Katsevich and Ramdas, 2020 calibration

Proposition 12

Let $\alpha \leq 0.31$. Then, in the top-k setting with p-values under the null independent,

$$\zeta_k := \left\lfloor \frac{\log(1/\alpha)}{\log(1+\log(1/\alpha))}\left(1 + mp_{(k)}\right) \right\rfloor$$

gives a reference family with JER *bounded by α.*

Proposition 13 (Uniform version of Wellner inequality, Meah et al., 2024)

Let $U_1, \ldots, U_m \overset{iid}{\sim} \mathcal{U}\big([0,1]\big)$ and $\kappa = \frac{\pi^2}{6}$. For all $\delta \in (0,1]$,

$$\mathbb{P}\left(\forall t \in [0,1], m^{-1}\sum_{1\leq i\leq m} \mathbb{1}_{U_i \leq t} \leq t h^{-1}\left(\frac{2\log(\kappa/\delta)+4\log\left(1+\log_2(1/t)\right)}{mt}\right)\right) \geq 1-\delta,$$

with $h(\lambda) = \lambda(\log\lambda - 1) + 1$.

Appendix: Confidence envelopes with heterogeneous data

Heterogeneous simultaneous k-FWER

Derive a simultaneous k-FWER inequality from a FDR control procedure

Example 8 (Simes inequality)

Let $\alpha \in [0,1]$. Assume that $(p_i)_{i\in\mathcal{H}_0}$ is a family of independent p-values. Then

$$\mathbb{P}\left(\exists i \in \{1,\dots,m_0\} : p_{(i:\mathcal{H}_0)} \leq \frac{\alpha i}{m}\right) \leq \alpha.$$

Derivation from BH procedure:

$$\begin{aligned}
\mathbb{P}\left(\exists i \in \mathcal{H}_0,\, p_{(i:\mathcal{H}_0)} \leq \frac{\alpha i}{m_0}\right) &= \mathbb{P}\left(\left|R^{\mathrm{BH}}\left((p_i)_{i\in\mathcal{H}_0}\right)\right| > 0\right) \\
&= \mathrm{FWER}\left(R^{\mathrm{BH}}\left((p_i)_{i\in\mathcal{H}_0}\right)\right) \text{ since each } i \in \mathcal{H}_0 \text{ is true,} \\
&= \mathrm{FDR}\left(R^{\mathrm{BH}}\left((p_i)_{i\in\mathcal{H}_0}\right)\right) \\
&\leq \alpha, \text{ by BH procedure.}
\end{aligned}$$

Implies $JER(\mathfrak{R}) \leq \alpha$ for $R_k = \left\{i : p_i \leq \frac{\alpha k}{m}\right\}$ and $\zeta_k = k-1$.

Step-up procedures

Definition 7

Let $\tau = (\tau_i)_{i \in [\![1,m]\!]}$ be a family of thresholds.
Then the step up procedure associated to τ return
$R^{\mathrm{SU}(\tau)} = \{i \in [\![1,m]\!] : p_i \leq \tau_{\hat{k}}\}$ *and* $\hat{k}^{\mathrm{SU}(\tau)} = \max\Big\{k : p_{(k)} \leq \tau_k\Big\}$.

The BH procedure is a **step-up** procedure.

A heterogeneous FDR control procedure

Theorem 6 (Döhler et al. (2018), Theorem 1)

Assume that $(p_i)_{i\in\mathcal{H}_0}$ is a family of independent p-values. Furthermore, assume that the two families $(p_i)_{i\in\mathcal{H}_0}$ and $(p_i)_{i\in\mathcal{H}_1}$ are independent. Let τ be a threshold family such that $\tau_m < 1$. Then

$$\mathrm{FDR}\left(R^{\mathrm{SU}(\tau)}\right) \leq \max_{1\leq k\leq m}\left\{\max_{\substack{A\subset[\![1,m]\!]\\|A|=m-k+1}}\left\{\frac{1}{k}\sum_{i\in A}\frac{F_i(\tau_k)}{1-F_i(\tau_m)}\right\}\right\}^{(\star)}.$$

Let, for all $(s,t)\in[0,1)\times[0,1]$,

$$\mathbf{H}(s,t) = \left(\frac{F_i(t)}{1-F_i(s)}\right)_{1\leq i\leq m},$$

for $i\in[\![1,m]\!]$, $\mathcal{A}_i$ be the support of p_i and

$$\mathcal{A} = \{0\}\bigcup_{i\in[\![1,m]\!]}\mathcal{A}_i.$$

Threshold family construction

Theorem 7 (Heterogeneous Simes-like inequality)

Let τ be a threshold family satisfying $(\star) \leq \alpha$, and assume that $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p-values. Then, the following inequality holds :

$$\mathbb{P}\Big(\exists i \in \mathcal{H}_0,\, p_{(i:\mathcal{H}_0)} \leq \tau_i\Big) \leq \alpha.$$

Example 9 (Heterogeneous step-up procedure)

Let $\alpha \in (0,1)$. Set

$$\tau_m \in \left\{ t \in \mathcal{A} : t < 1 \text{ and } \big(\mathbf{H}(t,t)\big)_{[1]} \leq m\alpha \right\},$$

and, for all $k \in [\![1, m-1]\!]$, let

$$\tau_k = \max\left\{ t \in \mathcal{A}, t \leq \tau_m : \big(\mathbf{H}(\tau_m,t)\big)_{[1]} + \cdots + \big(\mathbf{H}(\tau_m,t)\big)_{[m-k+1]} \leq k\alpha \right\}.$$

FDR control inequality for adaptivity

Proposition 14

From the proof of Theorem 6 in Döhler et al. (2018) the following slightly more refined result can be derived:

$$\mathrm{FDR}\left(R^{\mathrm{SU}(\tau)}\right) \leq \max_{1\leq k\leq m}\left\{\max_{\substack{A\subset[\![1,m]\!]\\|A|=m-k+1}}\left\{\frac{1}{k}\sum_{i\in A\cap\mathcal{H}_0}\frac{F_i(\tau_k)}{1-F_i(\tau_m)}\right\}\right\}^{(\star)}.$$

Example 10

Let $\alpha \in (0,1)$ and τ_m under the same constraints than in Example 9 and, for all $k \in [\![1, m-1]\!]$, let

$$\tilde{\tau}_k = \max\left\{t \in \mathcal{A}, t \leq \tilde{\tau}_m : \left(\mathbf{H}(\tilde{\tau}_m, t)\right)_{[1]} + \cdots + \left(\mathbf{H}(\tilde{\tau}_m, t)\right)_{[(m-k+1)\wedge m_0 : \mathcal{H}_0]} \leq k\alpha\right\}.$$

Then $\tilde{\tau}$ satisfies $(\star) \leq \alpha$.

Adaptive thresholds

Theorem 8 ($\hat{m}_0$-adaptivity of a threshold family)

Let $\tilde{\tau}$ be a threshold family such that $(\star) \leq \alpha$. Furthermore, assume that $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p-values. Then, for all (possibly data-dependant) threshold family τ such that $\tau_k \leq \tilde{\tau}_k$ for all $k \in [\![1, m]\!]$ under the event $\left\{\forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\right\}$, the following inequality holds :

$$\mathbb{P}\Big(\exists i \in \mathcal{H}_0,\, p_{(i:\mathcal{H}_0)} \leq \tau_i\Big) \leq \alpha.$$

Adaptive k-FWER inequality

Corollary 1 (Heterogeneous Simes like adaptive inequality)

Let $\tilde{\tau}$ be defined by Example 10 and $\hat{m}_0$ be a statistic such that

$$\left\{\forall i \in \mathcal{H}_0,\ p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\right\} \subset \{m_0 \leq \hat{m}_0\}.$$

Furthermore, assume that $(p_i)_{i\in\mathcal{H}_0}$ is a family of independent p-values. Then the thresholds family defined by $\tau_m = \tilde{\tau}_m$ and $\forall k \in [\![1, m-1]\!]$,

$$\tau_k = \max\left\{t \in \mathcal{A}, t \leq \tau_m : \left(\mathbf{H}(\tau_m, t)\right)_{[1]} + \cdots + \left(\mathbf{H}(\tau_m, t)\right)_{[(m-k+1)\wedge\hat{m}_0]} \leq k\alpha\right\}$$

verifies the inequality

$$\mathbb{P}\left(\exists i \in \mathcal{H}_0,\ p_{(i:\mathcal{H}_0)} \leq \tau_i\right) \leq \alpha.$$

Appendix: e-Closure principle with heterogeneous data

Appendix: e-Closure principle with heterogeneous data

Precisions on the shortcut

Shortcut for the heterogeneous e-Closure principle

To verify that a rejection set R belongs to $\mathcal{R}^{\mathrm{SC}}_{\alpha,g}(\mathbf{p})$, it suffices to check for all $a \leq |R|$ and $b \leq |R^c|$ that

$$\frac{\alpha}{a+b} \sum_{1 \leq i \leq a} g\left(\frac{1}{a+b} \left(\sum_{1 \leq j \leq a} \left(\mathbf{F}(p_{[i:R]}) \right)_{[j:R]} + \sum_{1 \leq j \leq b} \left(\mathbf{F}(p_{[i:R]}) \right)_{[j:R^c]} \right) \right) +$$

$$\frac{\alpha}{a+b} \sum_{1 \leq i \leq b} g\left(\frac{1}{a+b} \left(\sum_{1 \leq j \leq a} \left(\mathbf{F}(p_{[i:R^c]}) \right)_{[j:R]} + \sum_{1 \leq j \leq b} \left(\mathbf{F}(p_{[i:R^c]}) \right)_{[j:R^c]} \right) \right)$$

$$\geq \frac{a}{|R|}$$